Ordinary PDK LVS. Each PDK ships its deck, and neither changes when the part becomes part of an assembly.

LEVEL 1

Per-die LVS

shipped by each die PDK

Does each die match its own schematic?

Die schematic netlist
devices, terminals, nets

→

Netlist compare

←

Die layout (GDS)
device extraction from mask layers

LEVEL 2

Interposer LVS

shipped by the interposer PDK; not yet run on an exported interposer

Does the interposer match its own netlist?

Interposer netlist
nets and pad names

→

Netlist compare

←

Interposer layout (GDS)
extraction over the metal and via stack

The level the assembly adds, checked by assembly-lvs.

LEVEL 3

Assembly LVS

checked by assembly-lvs

Are the dies connected to each other as the system intends?

Declared nets
die pins joined by each net, carried in the assembly description

→

Connectivity-graph reconciliation

←

Placed layout set
die layouts and interposer layout, with placement, orientation and mirroring applied

No device spans a die boundary, so a device-graph comparator has nothing to anchor the inter-die question on.